Chapter 3
The Diffusion Equation

3.1 Diffusion Equation

One dimensional diffusion equation can be written as

$$\frac{\partial \phi}{\partial t} = \alpha \frac{\partial^2 \phi}{\partial x^2} \tag{3.1}$$

the dependent variable ϕ (such as temperature, species, momentum, etc.) diffuses in an infinite medium in both directions (to the left and right, x^+ and x^-) without any preference. The rate of diffusion depends on the parameter α, where α stands for thermal diffusion coefficient, mass diffusion coefficient or kinematics viscosity, for energy, mass, and momentum diffusion, respectively. The diffusion process becomes faster as the parameter α increases. Order of magnitude analysis yields,

$$\frac{1}{\tau} \approx \alpha \frac{1}{\ell^2} \tag{3.2}$$

where τ and l are time and length scales, respectively.

For a given time, the scalar ϕ diffusive to distance l is proportional to

$$\ell \approx \sqrt{\alpha\tau} \tag{3.3}$$

Equation 3.1 has second derivative in space, therefore, the diffusion takes place in both directions and requires two boundary conditions. Also, Eq. 3.1 has a first derivative in time, the diffusion is one directional in time, in other words, the diffusion at any point depends on the previous time and no information can be transferred from the future time. Also, it requires an initial condition to solve Eq. 3.1.

For instance, if a drop of ink is added to a glass filled with water, the ink diffuses in all directions and concentration of the ink at any spot in the water decreases as the time proceeds until it reaches the equilibrium condition.

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The concentration of ink at any location depends on the pervious time and on the concentration at the boundary of that spot.

3.1.1 Example 3.1

One-dimensional heat diffusion equation can be written as,

$$\rho C \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k \frac{\partial T}{\partial x}\right) \tag{3.4}$$

where T is temperature, ρ, C and k are density, specific heat and thermal conductivity of the medium, respectively. The heat diffusion due to molecular action, depends on thermal conductivity (k), density (ρ) and specific heat (C). For a constant k, the above equation can be written as

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \tag{3.5}$$

where α is the thermal diffusivity ($k/\rho C$). Hence, the only parameter that controls the diffusion of heat is the thermal diffusivity of the medium, which is a property of the material.

3.1.2 Example 3.2

One-dimensional momentum diffusion can be written as,

$$\frac{\partial u}{\partial t} = v \frac{\partial^2 u}{\partial y^2} \tag{3.6}$$

where u is velocity, v is kinematic viscosity of the fluid. For fluid between parallel plates (Fig. 3.1) if the upper plate is set to a motion, the momentum diffuses according to the above equation. The rate of momentum diffusion depends on the

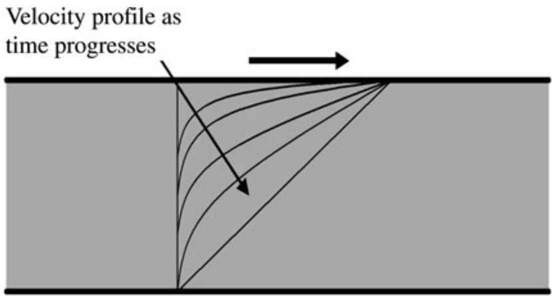


Fig. 3.1 Momentum diffusion in a fluid confined between two parallel plates due to the motion of the upper lid

viscosity of the fluid. The time needed for the bottom plate to sense the motion of the upper plate depends on the viscosity of the fluid (high viscosity means less sensation time), and the distance between the plates (Eq. 3.3).

3.2 Finite Differences Approximation

In the following section, numerical solution using finite difference approximation for one dimensional diffusion problem will be discussed in detail. Extending the method of solution to two and three dimensional problems is a straightforward procedure.

The main objectives of the approximation of the diffusion equation using the finite difference are two folds; first to show the difference and similarity between finite difference approximation and lattice Boltzmann method (LBM); also, to compare the results of the two methods.

As a first step, the domain need to be divided (discretized) into equal segments (it is not necessary for the length of each segment to be equal, however for simplicity we use equal segments) as shown in Fig. 3.2. The nodes are labeled as $0, 1, 2, 3, \ldots, i-2, i-1, i, i+1, i+2, \ldots n$.

An explicit finite difference approach can be used, forward in time and central differences in space. Approximating the diffusion equation at a node i, yields,

$$\frac{T_i^{n+1} - T_i^n}{\Delta t} = \alpha \frac{T_{i+1}^n - 2T_i^n + T_{i-1}^n}{\Delta x^2} \tag{3.7}$$

The above equation can be rearranged as:

$$T_i^{n+1} = T_i^n + \frac{\alpha \Delta t}{\Delta x^2}\left(T_{i+1}^n - 2T_i^n + T_{i+1}^n\right) \tag{3.8}$$

Equation 3.8 can be reformulated as,

$$T_i^{n+1} = T_i^n\left(1 - \frac{2\alpha\Delta t}{\Delta x^2}\right) + \frac{2\alpha\Delta t}{\Delta x^2}\left(\frac{T_{i+1}^n + T_{i+1}^n}{2}\right) \tag{3.9}$$

Let

$$\omega = \frac{2\alpha\Delta t}{\Delta x^2} \tag{3.10}$$

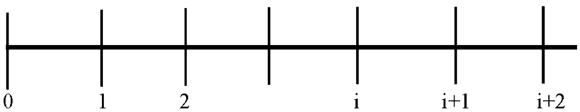


Fig. 3.2 One dimensional node distribution

then Eq. 3.9 can be written as

$$T_i^{n+1} = T_i^n(1-\omega) + \omega\left(0.5T_{i+1}^n + 0.5T_{i-1}^n\right) \tag{3.11}$$

For stability condition, the coefficients of the right-hand side terms must be positive. Hence, the term $(1-\omega)$ must be greater than or equal to zero, which implies that

$$\Delta t \le \frac{\Delta x^2}{2\alpha}$$

The finite difference approximation for the diffusion equation is manipulation in the form of Eq. 3.11 to compare the finite difference formulation with LBM.

Let us examine Eq. 3.11, the last term $(0.5[T_{i+1}^n + T_{i-1}^n])$ is an average of temperatures around T_i. In other words, the average term represents the equilibrium value of T_i. Since, T_i is at mid-distance from T_{i+1} and T_{i-1}, then at the equilibrium state, the value of T_i should be the average of it's neighbors values. It is appropriate to write the term $0.5[T_{i+1}^n + T_{i-1}^n]$ as T_i^{eq}, then Eq. 3.11, can be rewritten as,

$$T_i^{n+1} = T_i^n(1-\omega) + \omega T_i^{\mathrm{eq}} \tag{3.12}$$

In the next section we can see that Eq. 3.12 resembles LB equation (3.18).

3.3 The Lattice Boltzmann Method

The kinetic equation for the distribution function (temperature distribution, species distributions, 12, etc.), $f_k(x,t)$ can be written as:

$$\frac{\partial f_k(x,t)}{\partial t} + c_k \cdot \frac{\partial f_k(x,t)}{\partial x} = \Omega_k \tag{3.13}$$

$k = 1, 2$ (for one dimensional problem, D1Q2).

The left-hand side terms represent the streaming process, where the distribution function streams (advects) along the lattice link with velocity

$$c_k = \frac{\Delta x}{\Delta t} \tag{3.14}$$

The right-hand term, Ω_k, represents the rate of change of distribution function, f_k, in the collision process.

BGK approximation for the collision operator can be approximated as,

$$\Omega_k = -\frac{1}{\tau}\left[f_k(x,t) - f_k^{\mathrm{eq}}(x,t)\right] \tag{3.15}$$

The term τ represents a relaxation time toward the equilibrium distribution (f_k^{eq}), which is related to the diffusion coefficient on the macroscopic scale. The relation will be discussed latter on.

The kinetic LB equation (3.13) with BGK approximation can be discretized as,

$$\frac{f_k(x, t + \Delta t) - f_k(x, t)}{\Delta t} + c_k \cdot \frac{f_k(x + \Delta x, t + \Delta t) - f_k(x, t + \Delta t)}{\Delta x} = -\frac{1}{\tau}\left[f_k(x, t) - f_k^{\mathrm{eq}}(x, t)\right] \tag{3.16}$$

Note that $\Delta x = c_k \Delta t$, substituting in Eq. 3.16, which can be simplified to:

$$f_k(x + \Delta x, t + \Delta t) - f_k(x, t) = -\frac{\Delta t}{\tau}\left[f_k(x, t) - f_k^{\mathrm{eq}}(x, t)\right] \tag{3.17}$$

The above equation is the working horse for the diffusion problem in one dimensional space, which can be reformulated as,

$$f_k(x + \Delta x, t + \Delta t) = f_k(x, t)[1 - \omega] + \omega f_k^{\mathrm{eq}}(x, t) \tag{3.18}$$

where

$$\omega = \frac{\Delta t}{\tau}$$

is called a relaxation time.

It is interesting to note that the above Eq. 3.17 is similar to finite difference equation (3.12).

Equation 3.17 represents a number of equations for different k values ($k = 1$ and 2), in each direction.

Figure 3.3 shows schematic diagram for three nodes with necessary linkages, central and neighboring nodes, where $c_1 = c, c_2 = -c$. Figure 3.4 shows lattices distribution for 1-D, D1Q2.

The dependent variable in Eq. 3.1, ϕ, can be related to the distribution function f_i, as,

$$\phi(x, t) = \sum_{k=1}^{2} f_k(x, t) \tag{3.19}$$

The equilibrium distribution function f_k^{eq} (see Eq. 2.19), can be chosen as

$$f_k^{\mathrm{eq}} = w_k \phi(x, t) \tag{3.20}$$

where w_k stands for the weighting factor in the direction k. The weighting factor should satisfy the following criterion,

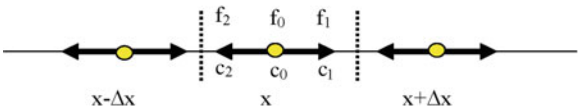


Fig. 3.3 Schematic diagram for 1-D lattice arrangement

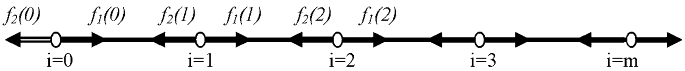


Fig. 3.4 Lattices for 1-D, diffusion problem

$$\sum_{k=1}^{2} w_k = 1 \tag{3.21}$$

The distribution function $f_k^{\mathrm{eq}}(x,t)$ defined in Eq. 3.20 can be summed along all directions and yields,

$$\sum_{i=0}^{2} f_k^{\mathrm{eq}}(x,t) = \sum_{i=0}^{2} w_k \phi(x,t) = \phi(x,t) \tag{3.22}$$

The relation between α and ω can be deduced from multi-scale expansion by using Chapman–Enskog expansion, which yields

$$\alpha = \frac{\Delta x^2}{\Delta t D}\left(\frac{1}{\omega} - \frac{1}{2}\right) \tag{3.23}$$

D is dimension of the problem ($D = 1$ for one dimensional problem, $D = 2$ for two dimensional problem and $D = 3$ for three dimensional problem). The weight factor for one-dimensional problem is $w_k = 1/2$, for $k = 1$ and 2.

In the following section Chapman–Enskog expansion will be illustrated for one dimensional diffusion equation. We will show how the relationship between diffusion coefficient and relaxation time can be established, step by step.

3.4 Equilibrium Distribution Function

For diffusion problem, it is appropriate that equilibrium distribution functions be assumed constants, where no macroscopic velocity is involved. Let,

$$f_i^{\mathrm{eq}} = A_i \tag{3.24}$$

Equilibrium distribution function must satisfy the conservation of mass and momentum,

$$\sum_{i=1}^{2} f_i^{\mathrm{eq}} = \Theta \tag{3.25}$$

and

$$\sum_{i=1}^{2} f_i^{\mathrm{eq}} c_i = 0 \tag{3.26}$$

Hence,

$$\sum_{i=1}^{2} A_i = A_1 + A_2 = \Theta \tag{3.27}$$

Also,

$$\sum_{i=1}^{2} A_i c_i = A_1 c_1 + A_2 c_2 = A_1 - A_2 = 0 \tag{3.28}$$

where $c_1 = 1$ and $c_2 = -1$. Hence $A_1 = A_2$ and from Eq. 3.51, $A_1 = \frac{\Theta}{2}$. In general,

$$A_i = \omega_i \Theta \tag{3.29}$$

Hence,

$$f_i^{\mathrm{eq}} = \omega_i \Theta \tag{3.30}$$

In the following sections, the relationship between LBM and macro-scale system will be discussed. In other words, the relationship between diffusion coefficient and LBM relaxation time will be established by the multi-scale analysis.

3.5 Chapman–Enskog Expansion

One dimensional diffusion equation in Cartesian coordinate can be written as,

$$\frac{\partial T(x,t)}{\partial t} = \Gamma \frac{\partial^2 T(x,t)}{\partial x^2} \tag{3.31}$$

where $T(x,t)$ is the dependent parameter (for instance, temperature for heat diffusion, velocity for momentum diffusion, species concentration for mass diffusion). It is assumed that diffusion coefficient, Γ, is a constant. The diffusion equation can be scaled spatially as, x is set to x/ϵ, where ϵ is a small parameter (Knudson number). Introducing the scale to diffusion equation yields,

$$\frac{\partial T(x,t)}{\partial t} = \Gamma \frac{\epsilon^2 \partial^2 T(x,t)}{\partial x^2} \tag{3.32}$$

Hence, in order to keep both sides of the equation at the same order of magnitude, the time t, must be scaled by $1/\epsilon^2$, i.e., t should be set to t/ϵ^2. For one-dimensional problem with two lattice velocity components ($c_i, i = 1, 2$), the temperature can be expressed as,

$$T(x,t) = \sum_{i=1}^{i=2} f_i(x,t) = f_1(x,t) + f_2(x,t) \tag{3.33}$$

From now and on, we write f_i instead of $f_i(x,t)$ for simplicity. Since the diffusion equation is linear, it is fair to assume that equilibrium distribution function is related to T (see pervious section for details) as,

$$f_i^{\mathrm{eq}} = w_i T(x,t) \tag{3.34}$$

By taking summation of both sides of the above equation,

$$\sum_i f_i^{\mathrm{eq}} = T = w_1 T + w_2 T \tag{3.35}$$

hence, $w_1 + w_2 = 1$, i.e., $w_1 = w_2 = \frac{1}{2}$, then,

$$f_1^{\mathrm{eq}} = w_1 T(x,t) \quad \text{and} \quad f_2^{\mathrm{eq}} = w_2 T(x,t) \tag{3.36}$$

where the weight factors must satisfy the relation

$$\sum_{i=1}^{i=2} w_i = 1$$

The distribution function can be expanded in terms of a small parameter, ϵ as

$$f_i(x,t) = f_i^0 + \epsilon f_i^1 + \epsilon^2 f_i^2 + \cdots \tag{3.37}$$

where f_i^0 is distribution function at the equilibrium conditions, equal to f_i^{eq} By summing above equation, it leads to,

$$\sum_{i=1}^{i=2} f_i(x,t) = \sum_{i=1}^{i=2} [f_i^0 + \epsilon f_i^1 + \epsilon^2 f_i^2 + \cdots] \tag{3.38}$$

Since, $f_1 + f_2 = T(x,t)$ and $f_1^0 + f_2^0 = T(x,t)$, hence other expanded term in the above equation should be zero, i.e.,

$$\sum_{i=1}^{i=2} f_i^1 = 0 \tag{3.39}$$

$$\sum_{i=1}^{i=2} f_i^2 = 0 \tag{3.40}$$

The updated distribution function $f_i(x + c_i\Delta t, t + \Delta t)$ is expanded using Taylor series,

$$f_i(x + c_i\Delta t, t + \Delta t) = f_i(x,t) + \frac{\partial f_i}{\partial t}\Delta t + \frac{\partial f_i}{\partial x}c_i\Delta t + 1/2\Delta t^2\left(\frac{\partial^2 f_i}{\partial t^2} + 2\frac{\partial^2 f_i}{\partial t\partial x}c_i + \frac{\partial^2 f_i}{\partial x^2}c_ic_i\right) + O(\Delta t)^3 \quad (3.41)$$

Introducing scaling for the above equation, i.e., $\frac{1}{\partial t}$ is replaced by $\frac{\epsilon^2}{\partial t}$ and $\frac{1}{\partial x}$ is replaced by $\frac{\epsilon}{\partial x}$, yields:

$$f_i(x + c_i\Delta t, t + \Delta t) = f_i(x,t) + \epsilon^2\frac{\partial f_i}{\partial t}\Delta t + \epsilon\frac{\partial f_i}{\partial x}c_i\Delta t + 1/2\Delta t^2\left(\frac{\epsilon^4\partial^2 f_i}{\partial t^2} + 2\frac{\epsilon^3\partial^2 f_i}{\partial t\partial x}c_i + \frac{\epsilon^2\partial^2 f_i}{\partial x^2}c_ic_i\right) + O(\Delta t)^3 \quad (3.42)$$

Lattice Boltzmann equation can be written as,

$$f_i(x + c_i\Delta t, t + \Delta t) = f_i(x,t) + \frac{\Delta t}{\tau}[f_i^0(x,t) - f_i(x,t)] \quad (3.43)$$

Substituting Eq. 3.42 into above equation and retaining terms up to order of ϵ^2 , yields:

$$\frac{1}{\tau}[f_i^0(x,t) - f_i(x,t)] = \epsilon^2\frac{\partial f_i}{\partial t} + \epsilon\frac{\partial f_i}{\partial x}c_i + 1/2\epsilon^2\Delta t\left(\frac{\partial^2 f_i}{\partial x^2}c_ic_i\right) + O(\Delta t)^2 + O(\epsilon^3) \quad (3.44)$$

where f_i^0 is f_i^{eq}. Expanding $f_i = f_i^0 + \epsilon f_i^1 + \epsilon^2 f_i^2 + \cdots$ and substituting in the above equation, yields,

$$-\frac{1}{\tau}(\epsilon f_i^1 + \epsilon^2 f_i^2 + \cdots) = \epsilon^2\frac{\partial f_i^0}{\partial t} + \epsilon\frac{\partial f_i^0}{\partial x}c_i + \epsilon^2\frac{\partial f_i^1}{\partial x}c_i + \frac{\Delta t}{2}\epsilon^2\frac{\partial f_i^0}{\partial x}c_ic_i + O(\epsilon^3) +)(\Delta t^2) \quad (3.45)$$

Equating term from both sides of the above equation of same order of ϵ:

Terms order of ϵ:

$$-\frac{1}{\tau}f_i^1 = \frac{\partial f_i^0}{\partial x}c_i \quad (3.46)$$

Latter on we may need derivative of f_i^1 with respect of x, hence by taking the derivative of above equation, gives:

$$-\frac{1}{\tau}\frac{\partial f_i^1}{\partial x} = \frac{\partial^2 f_i^0}{\partial x^2}c_i \quad (3.47)$$

Terms order of ϵ^2:

$$-\frac{1}{\tau}f_i^2 = \frac{\partial f_i^0}{\partial t} + \frac{\partial f_i^1}{\partial x}c_i + \frac{\Delta t}{2}\frac{\partial^2 f_i^0}{\partial x^2}c_i c_i \tag{3.48}$$

Substituting the derivative of f_i^1 with respect of x in the above equation, yields,

$$-\frac{1}{\tau}f_i^2 = \frac{\partial f_i^0}{\partial t} - \tau\frac{\partial^2 f_i^0}{\partial x^2}c_i c_i + \frac{\Delta t}{2}\frac{\partial^2 f_i^0}{\partial x^2}c_i c_i \tag{3.49}$$

To recover continuum diffusion equation (1.1), the above equation need to be summed over all states, $i = 1, 2$.

$$\sum_i\left[-\frac{1}{\tau}f_i^2\right] = \sum_i\left[\frac{\partial f_i^0}{\partial t} - \tau\frac{\partial^2 f_i^0}{\partial x^2}c_i c_i + \frac{\Delta t}{2}\frac{\partial^2 f_i^0}{\partial x^2}c_i c_i\right] \tag{3.50}$$

The first term on the left-hand side (LHS) is zero. The first term on the right-hand side (RHS), will be

$$\sum_i\frac{\partial f_i}{\partial t} = \frac{\partial\sum_i f_i}{\partial t} = \frac{\partial\Theta}{\partial t} \tag{3.51}$$

The term of,

$$\sum_i\left[\frac{\partial^2 f_i^0}{\partial x^2}c_i c_i\right] = \frac{\partial^2}{\partial x^2}\left(\sum_i[f_i^0 c_i c_i]\right) = \frac{\partial^2\Theta}{\partial x^2} \tag{3.52}$$

Hence Eq. 3.50 can be simplified as,

$$0 = \frac{\partial\Theta}{\partial t} - \left(\tau - \frac{\Delta t}{2}\right)\frac{\partial^2\Theta}{\partial x^2} \tag{3.53}$$

By comparing the above equation with continuum diffusion equation, the relaxation parameter, τ, can be related to the diffusion coefficient as,

$$\Gamma = \tau - \frac{\Delta t}{2} \tag{3.54}$$

Hence, a relationship is established between macro and meso-scales.

3.5.1 Normalizing and Scaling

It is more appropriate to solve specific problems in non-dimensionalized (normalized) forms rather than in dimensional form for a few reasons, the controlling parameters can be grouped into a few non-dimensional parameters, hence the results can be presented in more general forms. Also, the scaling from continuum mechanics (macro-scale) to LBM (meso-scale) becomes clear. For example, Eq. 3.1 is dimensional form for diffusion equation, α has units of m^2/s, x has unit of m and ϕ has units of °C, or K, or other units depending on what ϕ represents. However, let's assume that Eq. 3.1 represents heat diffusion equation, then ϕ represents temperature and has units of centigrade (°C). We can scale the temperature by $\theta = \frac{\phi - \phi_c}{\phi_h - \phi_c}$, where ϕ_c and ϕ_h are cold and hot temperatures binding the system. The x can be scaled by the length of the system L, i.e., nondimensionalized $x^* = x/L$. The nondimensional heat diffusion equation can be written as,

$$\frac{\partial \theta}{\partial t^*} = \frac{\partial^2 \theta}{\partial x^{*2}} \tag{3.55}$$

where t^* is non-dimensional time ($t^* = \alpha t/L^2$). The equation is independent of $\alpha; \theta$ and x ranges from 0 to 1.0, regardless of the range of the dimensional equation or the physical material of the system. More important, the equation can be solved easily with LBM considering the following scaling. In conventional numerical technique, say finite difference method (FDM) for example, we discretize the space domain into nodes and the distance between two nodes is Δx, and time domain also discretizes into Δt. If the domain descritzed into 100 nodes $(0, 1, 2, \ldots, 99, 100)$, then $\Delta x = 0.01$. Let us set Δt to 0.0001. Note that, Δt and Δx must be properly chosen to ensure stable solution if an explicit method is used. In LBM, the domain need to discretized also, however, the most popular method is to use unity for Δx and for Δt, i.e., $\Delta x = 1$ and $\Delta t = 1$. The question is what will be the number of lattices and time steps for total dimensional time of 5 (for example). We can select any total number of lattices, for example $N = 500$, then the nondimensional distance, x, at any site (i), is i/500. For instance site number 50, the x is 50/500=0.1. Time scale need to be worked out. Remember that t^* is equal to $\alpha t/L^2$ which in lattice unit should equal to $T\Delta t/\tau N^2$ where τ is relaxation time in LBM, T is total time and N is the total number of lattice sites. For the given example, we are interested in the results at dimensionless time of $t^* = 2.0$, hence,

$$2.0 = \frac{T\Delta t}{\tau N^2} \tag{3.56}$$

Since we select $N = 500, \Delta t = 1.0$, then have two free variables (τ and T), where T is total time steps in LBM units. If we select $\tau = 0.5\,\mathrm{m}$ then $T = 250,000$. Hence, the calculations should be carried out for 250,000 times to represent the results of 2.0 nondimensional time. For FDM, the number of time steps is 2/0.0001=20,000. In general, LBM needs less computer time than the FDM for the diffusion problems.

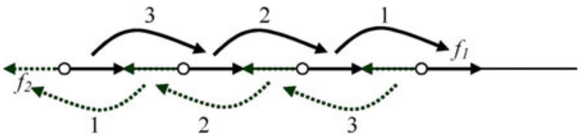


Fig. 3.5 Algorithm for streaming processes

3.5.2 Heat Diffusion in an Infinite Slab Subjected to a Constant Temperature

A slab initially at temperature equal to zero, $T = 0.0$. For time $t \geqslant 0$, the left surface of the slab is subjected to a high temperature and equal to unity, $T = 1.0$. The slab length is 100 units. Calculate the temperature distribution in the slab for $t = 200$. Compare the results of both methods, LBM and FDM, for $\alpha = 0.25$.

Solution Let us divide the domain of integration into $\Delta x = 1.0$ and $\Delta t = 1.0$.

Equation 3.23, can be used to calculate $\omega : 0.25 = (1/\omega - 0.5)$, then $\omega = 1.333333$.

The temperature is a function of x and t. Equation 3.19 states that

$$T(x,t) = \sum_{k=1}^{2} f_k(x,t) \tag{3.57}$$

and from Eq. 3.20, $f_k^{eq}(x,t) = w_k T(x,t)$, hence $f_1^{eq}(x,t) = 0.5T(x,t)$ and $f_2^{eq}(x,t) = 0.5T(x,t)$.

LBM consists of two steps, collision and streaming.

The collision step is:

$$f_k(x, t + \Delta t) = f_k(x,t)[1 - \omega] + \omega f_k^{eq}(x,t)$$

where $k = 1$ and 2.

The streaming step is:

$$f_k(x + \Delta x, t + \Delta t) = f_k(x, t + \Delta t).$$

In order not to save the pervious values of $f_k(x,t)$, and to avoid over writing updated values, the stream (updating) for $f_1(x,t)$ should be from $i = m$ to $i = 0$, while stream (updating) for $f_2(x,t)$ should be from $i = 0$ to $i = m$, as shown in Fig. 3.5.

As illustrated in Fig. 3.5, f_1 should be swept backward $(m, m-1, \ldots, 4, 3, 2, 1, 0)$ in programming, $i = m$, 0, -1. While, f_2 should be swept forward $(1, 2, 3 \ldots .)$ in programming, $i = 0$, m. By this procedure the overwriting of the updated values by old values can be avoided.

3.5.3 Boundary Conditions

At $x = 0$, the left-hand boundary condition, the value of f_2 can be obtained from the streaming process, then the unknown is only f_1 at the left boundary. Hence there is a need for an equation to relate f_1 to the known parameters. On the right side, $x = L$, the value of f_2 need to be known, while f_1 can be obtained from streaming process.

The following sections discuss treatment of different boundary conditions.

3.5.3.1 Constant Temperature Boundary Condition, Dirichlet Boundary Condition

The detailed flux balance at the boundary, $x = 0$, for D1Q2 is as follows,

$$f_1^{eq}(0,t) - f_1(0,t) + f_2^{eq}(0,t) - f_2(0,t) = 0 \tag{3.58}$$

and,

$$f_1^{eq}(0,t) = w_1 T_w = 0.5T_w \quad \text{and} \quad f_2^{eq}(0,t) = w_2 T_w = 0.5T_w.$$

Therefore at $x = 0.$, $f_1(0) + f_2(0) = T_w$, and from the streaming processes, $f_2(0) = f_2(1)$, then $f_1(0)$ can be determined as, $f_1(0) = T_w - f_2(0)$. Figure 3.6 shows the comparison of predicted results obtained by LBM and FDM for a slab with a constant temperature boundary condition.

3.5.3.2 Adiabatic Boundary Condition, Zero Flux Condition

The temperature gradient is zero, which implies that $T(m) = T(m-1)$.

Hence,

$$f_1(m) + f_2(m) = f_1(m-1) + f_2(m-1),$$

or

$$f_1(m) = f_1(m-1) \quad \text{and} \quad f_2(m) = f_2(m-1),$$

where m denotes the last lattice node.

3.5.4 Constant Heat Flux Example

Same example as before, except that the left-hand boundary is exposed to a constant heat flux of $100\,\mathrm{W/m^2}$ instead of a constant temperature. The thermal conductivity of the slab is 20 W/mK (Figs. 3.6 and 3.7).

Solution The procedure is same as for previous example, except that we need to determine $f_1(0)$ at $x = 0$.

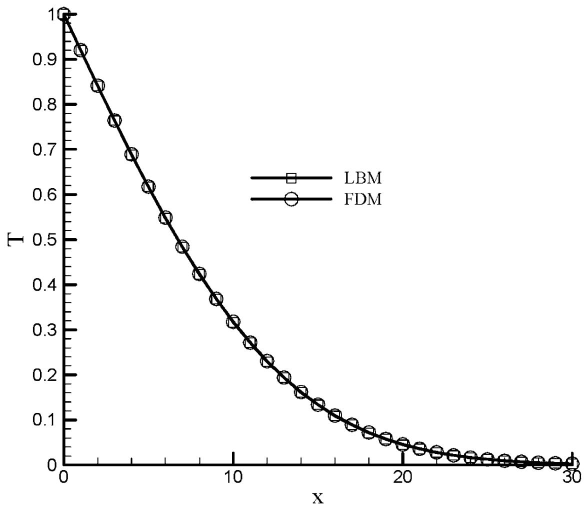


Fig. 3.6 Shows the comparison of predicted results obtained by LBM and FDM, for a slab with a constant temperature problem

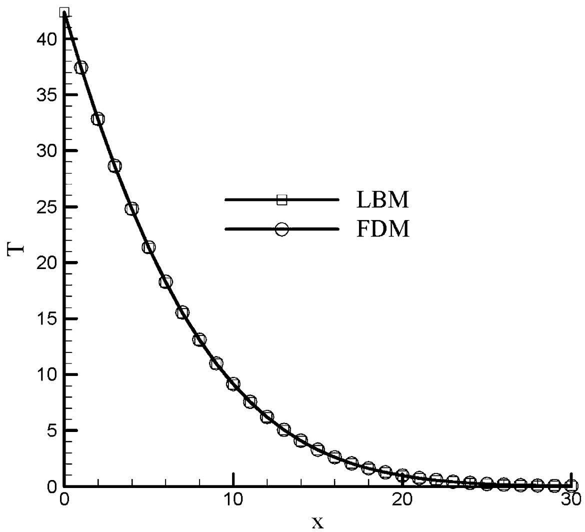


Fig. 3.7 Shows a comparison of results predicted by LBM and FDM for a slab subjected to a constant heat flux at the boundary

Fourier Law states that the heat flux is proportional to the temperature gradient:

$$q'' = -k\frac{\partial T}{\partial x} \tag{3.59}$$

where, k is the thermal conductivity of the material.

The above equation can be approximated at $x = 0$ by finite difference method, as:

$$q'' = -k\frac{T(1) - T(0)}{dx} \tag{3.60}$$

Substituting $T(1) = f_1(1) + f_2(1)$ and $T(0) = f_1(0) + f_2(0)$ in the above equation and solving for $f_1(0)$ gives,

$$f_1(0) = f_1(1) + f_2(1) - f_2(0) + \frac{q''dx}{k} \tag{3.61}$$

Hence $f_1(0)$ can be calculated from above equation, where $f_2(0)$ can be obtained from streaming step as before. The computer code is given in the Appendix.

Same computer code as discussed before (see Appendix for codes), except the boundary condition should be replaces by:

```
! Boundary condition

f1(0)=f1(1)+f2(1)-f2(0)+flux*dx/tk          ! constant heat flux boundary condition, x=0.

f1(m)=f1(m-1)                               ! adiabatic boundary condition, x=L

f2(m)=f2(m-1)                               ! adiabatic boundary condition, x=L
```

Figure 3.7 shows a comparison of results predicted by LBM and FDM for a slab subjected to a constant heat flux at the boundary.

3.6 Source or Sink Term

The source (sink) term can be manipulated as follows. Let us assume that we have a slab with a uniformly distributed heat source, such as a conductor subjected to a constant electric potential difference (voltage), or a chemical reaction taking place through a plane reactor. Assuming that the heat transfer is one-dimensional, then the differential equation for the such a problem can be written as,

$$\rho C \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k \frac{\partial T}{\partial x}\right) + q_g \tag{3.62}$$

where q_g is the rate of heat generation per unit volume. For a constant thermal conductivity, the equation can be rewritten as,

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} + \frac{q_g}{\rho C} \tag{3.63}$$

In Lattice Boltzmann, the source term can be added to the LB equation as

$$f_k(x, t + \Delta t) = f_k(x, t)[1 - \omega] + \omega f_k^{eq}(x, t) + \Delta t w_k S \tag{3.64}$$

where S is the force or source term, which can be related to q_g as

$$S = \frac{q_g}{\rho C} \tag{3.65}$$

notice that unit of S is unit of temperature (°C) per unit of time. The value of w_k is 0.5 for both w_1 and w_2.

The LBM code can be simply modified, by adding an extra term in calculating f_1 and f_2.

In the LBM codes, the following modifications need to be done,

```
======================================
      do i=0,m

         rho(i)=f1(i)+f2(i)
            source=qg/(rho(i)*cp)
         feq=0.5*rho(i)
f1(i)=omega*feq+(1.-omega)*f1(i)+dt*0.5*source

            f2(i)=omega*feq+(1.-omega)*f2(i)+dt*0.5*source
end do
=============================================
```

3.7 Axi-Symmetric Diffusion

One-dimensional radial diffusion in cylindrical or spherical geometries can be expressed as,

$$\frac{\partial T}{\partial t} = \frac{\alpha}{r^k}\frac{\partial}{\partial r}\left(r^k \frac{\partial T}{\partial r}\right) \tag{3.66}$$

where k equal to 1 and 2 for cylindrical and spherical coordinate systems, respectively. The above equation can be expanded as,

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial r^2} + \frac{k\alpha}{r}\frac{\partial T}{\partial r} \tag{3.67}$$

The above equation is similar to diffusion equation in Cartesian coordinate with an extra term, the last term, which can be treated as a source term.

Analytical solution for heat conduction (steady state) in cylindrical shell without heat generation can be written as,

$$\theta = \frac{\ln(r/r_{\mathrm{i}})}{\ln(r_{\mathrm{o}}/r_{\mathrm{i}})} \tag{3.68}$$

where $\theta = \frac{(T-T_{\mathrm{i}})}{(T_{\mathrm{o}}-T_{\mathrm{i}})}$, T_{i} and T_{o} are temperature at r_{i} (inner radius) and r_{o} (outer radius), respectively.

BGK Lattice Boltzmann equation with source term can be written as Eq. 3.64 with $\omega = \frac{\Delta t}{\alpha + 0.5}$. The source term S can be treated either

$$\frac{\alpha n}{r}\frac{T_{j+1} - T_{j-1}}{2\Delta r_j} \tag{3.69}$$

or

$$\frac{\alpha n}{r}\frac{f_{2,j+1} - f_{1,j-1}}{\tau} \tag{3.70}$$

where $\tau = \alpha + 0.5$.

For more details about solving axis-symmetric (cylindrical or spherical with or without heat generation) diffusion problem with LBM, we refer to paper by Mohamad (2009).

3.8 Two-Dimensional Diffusion Equation

The method of solution and equation for two and three dimensional problems are similar to the one dimensional problem. The steps of the solution are exactly the same, except that the number of streaming directions increase and the weighting factors for each direction should be evaluated. In the following section the algorithm of the solution will be discussed.

Three types of lattice arrangements will be discussed, namely D2Q4, D2Q5, and D2Q9. D2Q4 and D2Q5 are simple extension for 1-D problem, while D2Q9 is more involved. The main objective of introducing D2Q9 is for future analysis of fluid flow problems, Navier–Stokes equations. Hence, it aims to build a background for more complicated problems, mainly for fluid flow problems.

3.8.1 D2Q4

The common terminology used in LBM is to refer to the dimension of the problem and the number of speed is using D*n*Q*m*, where *n* represent the dimension of the problem (2 for 2-D and 3 for 3-D) and *m* refers to the speed model, as mentioned before.

Since the diffusion phenomena has no directional preference and only gradient determines the diffusion process, unlike advection process, where the information follows the direction of the flow. Full isotropy can be insured by 90° rotation of the coordinates; hence, it is sufficient to use only four streaming directions, Fig. 3.8.

Note that the numbering of *f* is arbitrary, you can use 1, 2, 3 and 4 instead of 1, 3, 2 and 4.

As usual, the LBM consists of two steps, collisions and streaming.

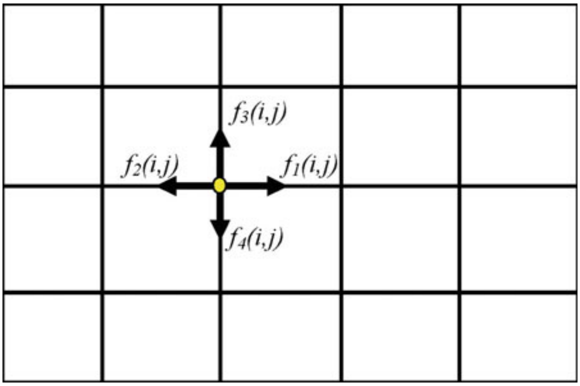


Fig. 3.8 Lattices for 2-D, diffusion problem

The collisions step without forcing function is:

$$f_k(x, y, t + \Delta t) = f_k(x, y, t)[1 - \omega] + \omega f_k^{\mathrm{eq}}(x, y, t) \tag{3.71}$$

$k = 1$, 2, 3 and 4.

The streaming step is:

$$f_k(x + \Delta x, y + \Delta y, t + \Delta t) = f_k(x, y, t + \Delta t) \tag{3.72}$$

$k = 1$, 2, 3 and 4.

In streaming step, for example $f_1(i,j)$ moves to $f_1(i+1,j)$, $f_2(i,j)$ moves to $f_2(i-1,j)$, $f_3(i,j)$ moves to $f_3(i,j+1)$ and $f_4(i,j)$ moves to $f_4(i,j-1)$. $w(1) = w(2) = w(3) = w(4) = 1/4$.

Indeed, nothing new compared with 1-D problem. In programming, one needs to take care of not over writing values of f's, other than that, every step is similar to the 1-D algorithm.

The $w_i = 0.25$ for $i = 1$, 2, 3 and 4. Also,

$$\alpha = \frac{\Delta x^2}{2\Delta t}\left(\frac{1}{\omega} - \frac{1}{2}\right) \tag{3.73}$$

3.8.2 D2Q5

Similar to D2Q4, except that there will be a particle residing at the center of the lattice (f_0). The weight coefficients are, 2/6 for f_0 and 1/6 for f_1,f_2,f_3 and f_4. The diffusion coefficient is,

$$\alpha = \frac{\Delta x^2}{3\Delta t}\left(\frac{1}{\omega} - \frac{1}{2}\right) \tag{3.74}$$

Other than that every step and algorithm is the same as for D2Q4, of course there is no streaming for the particle at the center of the lattice (f_0).

3.9 Boundary Conditions

In the following subsections, different boundary conditions will be discussed in detail.

3.9.1 The Value of the Function is Given at the Boundary

For example, temperature at the left and bottom boundaries are given, i.e., $T = C_1$at $x = 0$ and $T = C_2$ at $y = 0$, (C_1 and C_2 are constants), $f_1 + f_2 + f_3 + f_4 = T$. Then, the unknowns are f_1 and f_3 because f_2 and f_4 can be obtained from streaming step, i.e., $f_2(i,j) = f_2(i+1,j)$ and $f_4(i,j) = f_4(i,j+1) \cdot f_1$ and f_3 can be calculated by using flux conservation equation, i.e., $f_1^{eq} - f_1 + f_2^{eq} - f_2 = 0$, since $w = 0.25$ for all streaming directions and $f_1^{eq} = f_2^{eq} = 0.5C_1$, then $f_1 = 0.25C_1 + 0.25C_1 - f_2$, i.e., $f_1 = 0.5C_1 - f_2$ similarly $f_3 = 0.5C_2 - f_4$. This ensures the flux conservation, which conserve the physics of the problem.

However, if T is given on right boundary =, then f_2 need to be determined, similarly. The above argument is valid for both D2Q4 and D2Q5.

3.9.2 Adiabatic Boundary Conditions, for Instance

$$\frac{\partial T}{\partial x} = 0 \tag{3.75}$$

by using finite difference approach, $\frac{\partial T}{\partial x} = 0$, can be approximated as

$$\frac{T(i+1,j) - T(i,j)}{\Delta x} = 0 \tag{3.76}$$

(first order accurate). It can simplified as $T(i,j) = T(i+1,j)$. Hence,

$$\begin{aligned} f_1(i,j) + f_2(i,j) + f_3(i,j) + f_4(i,j) = {} & f_1(i+1,j) + f_2(i+1,j) \\ & + f_3(i+1,j) + f_4(i+1,j) \end{aligned} \tag{3.77}$$

It is rational to assume that $f_1(i,j) = f_1(i+1,j), f_2(i,j) = f_2(i+1,j)$ and so on.

3.9.3 Constant Flux Boundary Condition

For instance

$$k\frac{\partial T}{\partial x} = q \tag{3.78}$$

The finite difference approximation for the above boundary condition is,

$$k\frac{T(i+1,j) - T(i,j)}{\Delta x} = q \tag{3.79}$$

which can be rewritten as:

$$T(i,j) = T(i+1,j) - \frac{q\Delta x}{k} \tag{3.80}$$

therefore

$$f_1(i,j) = f_1(i+1,j) - \frac{q\Delta x}{k} \tag{3.81}$$

Other boundary conditions can be formulated similarly, i.e., using finite difference approach. It is also possible to use higher order approximations.

3.10 Two-Dimensional Heat Diffusion in a Plate

A two dimensional square slab shown in Fig. 3.9 is subjected to the given boundary conditions. Initially the slab was at zero temperature. For time $\geqslant 0$, the boundary at $x = 0$ is subjected to a high temperature of value 1.0 and other boundaries are kept as before. The length of the domain is 100 units. Determine the temperature distribution in the slab at time = 400 units (s). Compare the results obtained by using LB and FD methods. Thermal diffusivity is 0.25 and $xl = yl = 1.0$. The computer codes can be found in the Appendix.

Note that for stability conditions the time step for FDM is 0.2. There is no such a problem in using a time step of 1.0 with the LBM method. Hence, LBM is much faster and efficient than FDM.

3.10.1 D2Q9

For fluid flow problems (Navier–Stokes equation), where the advection term (nonlinear term) is important, the D2Q4 or D2Q5 model is not sufficient to insure

Fig. 3.9 A square (2-D) domain with coordinate system

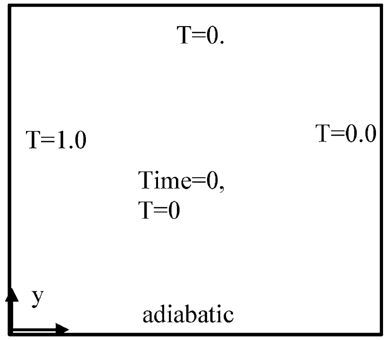
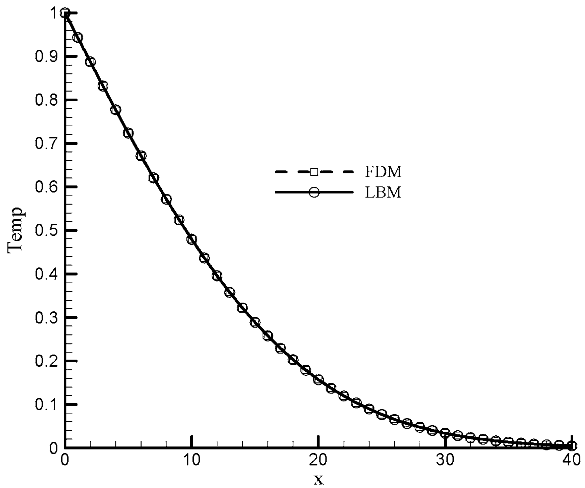


Fig. 3.10 Temperature distribution along the middle line ($y = 0.5$)



the macroscopic isotropy. It is common to use D2Q9. It is illustrative to introduce D2Q9 for a diffusion problem. The knowledge obtained in solving diffusion problem using D2Q9 can be easily extended for advection–diffusion and Navier–Stokes problems.

The macroscopic diffusion coefficient can be related to relaxation factor, ω, as

$$\alpha = \frac{\Delta x^2}{3\Delta t}\left(\frac{1}{\omega} - \frac{1}{2}\right) \tag{3.82}$$

Figures 3.10 and 3.11 compare results obtained by LBM and FDM for the given problem (Fig. 3.9). The domain of interest should be divided into lattices as shown in Fig. 3.12

The Solution Procedure Note that numbering of f is arbitrary. For simplicity of coding, it is more convenient to use numbering of streaming links as shown in Fig. 3.12

As usual, the LBM consists of two steps, collisions and streaming.

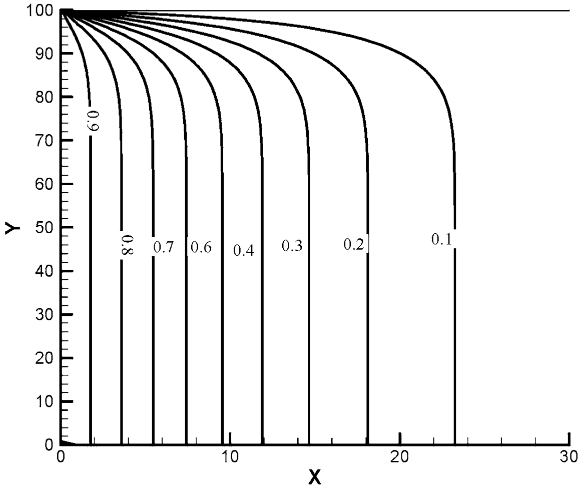


Fig. 3.11 Isotherm comparison between predictions of LBM and FDM, it is difficult to observe the difference between the predictions of both the methods

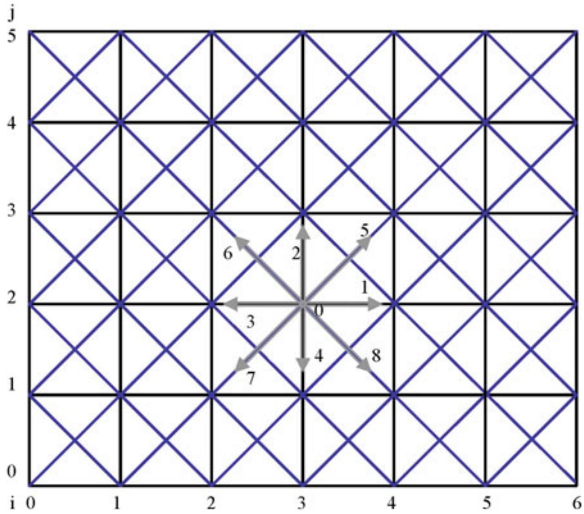


Fig. 3.12 Lattice arrangement for D2Q9

The collisions step without forcing function is:

$$f_k(x, y, t + \Delta t) = f_k(x, y, t)[1 - \omega] + \omega f_k^{\mathrm{eq}}(x, y, t) \tag{3.83}$$

$k = 0, \ldots, 8.$

The streaming step is:

$$f_k(x + \Delta x, y + \Delta y, t + \Delta t) = f_k(x, y, t + \Delta t) \tag{3.84}$$

$k = 1, \ldots, 8.$

In streaming process, for example $f_1(i,j)$ moves to $f_1(i+1,j)$,$f_2(i,j)$ moves to $f_2(i,j+1)$,$f_3(i,j)$ moves to $f_3(i-1,j)$,$f_4(i,j)$ moves to $f_4(i,j-1)$,$f_5(i,j)$

moves to $f_5(i+1,j+1)$, $f_6(i,j)$ moves to $f_6(i-1,j+1)$, $f_7(i,j)$ moves to $f_7(i-1,j-1)$ and $f_8(i,j)$ moves to $f_8(i+1,j-1)$. In programming, attention should be taken to not override new values in the streaming process.

The values of the weighting factors are:
$w(0) = 4/9$
$w(1) = w(2) = w(3) = w(4) = 1/9$ and
$w(5) = w(6) = w(7) = w(8)=1/36$.

The values of the streaming velocity along the lines are: $c_0(0,0), c_1(c,0), c_2(0,c), c_3(-c,0), c_4(0,-c), c_5(c,c), c_6(-c,c), c_7(-c,-c)$ and $c_8(c,-c)$ where $c = \Delta x/\Delta t$.

The initial distribution function $f_i(i,j)$ should be evaluated as,

$$f_k(i,j) = w(k)\phi(i,j) \tag{3.85}$$

The equilibrium function can be calculated using Eq. 3.20, $f_k^{\text{eq}}(i,j) = w(k)\phi(i,j)$ and $\phi(i,j)$ can be calculated as,

$$\phi(x,t) = \sum_{k=1}^{2} f_k(x,t) \tag{3.86}$$

3.10.2 Boundary Conditions

For Dirichlet boundary condition, the flux conservation principle will be used. For instance, the following procedure can be applied to evaluate f_5 at the boundary ($x = 0$), for a given ϕ value (ϕ_{wall}) at the left-hand boundary: $f_5^{\text{eq}} - f_5 + f_7^{\text{eq}} - f_7 = w_5\phi_{\text{wall}} + w_7\phi_{\text{wall}}$, f_7 is known from streaming process, then $f_5 = w_5\phi_{\text{wall}} + w_7\phi_{\text{wall}} - f_7$.

Similarly, the other unknown streaming functions can be evaluated. Adiabatic boundary condition can be treated as discussed before. For details see the computer code in appendix. The computer code is given in appendix for the same problem as discussed before for D2Q5, i.e., heat diffusion in a square plate.

3.10.3 Constant Flux Boundary Conditions

For constant heat flux boundary condition, for instance $x = 0$,

$$k\frac{\partial T}{\partial x} = q. \tag{3.87}$$

The above equation can be approximated using FDMs, which yields,

$$k\frac{T(i+1,j)-T(i,j)}{\Delta x}=q \tag{3.88}$$

At $x=0$, the equation can be written as

$$k\frac{T(1,j)-T(0,j)}{\Delta x}=q \tag{3.89}$$

which can be rearranged as,

$$T(0,j)=T(1,j)+\frac{q\Delta x}{k} \tag{3.90}$$

Then follow similar procedure as for a constant temperature problem condition with an extra term.

3.11 Problems

1. Extend the LBM for three dimensional problem, solve heat diffusion in a cubic material subjected to a hot vertical wall on one face and other faces kept at constant but cold temperature. Compare finite difference solution with LBM solution at dimensionless time of 40 and after steady state reaches.

2. Solve one dimensional problem with convective boundary condition.

3. Solve diffusion problem for the plate shown in Fig. 3.13, using LBM. The plate left-hand vertical boundary maintained at $T=1.0$ and right-hand vertical boundary at $T=0.0$. The other boundaries are assumed to be adiabatic. Plot steady state isotherms and compare it with FDM.

4. In certain applications, the thermal conductivity of the material may be a function of temperature. For instance $\alpha=0.25+0.5T^2$, α, is the thermal diffusivity of the material and T is dimensionless temperature. Solve heat diffusion equation in such a material. The domain of the solution is 2 units in length and initial was at $T=0$, suddenly the temperature of left-hand boundary is set to

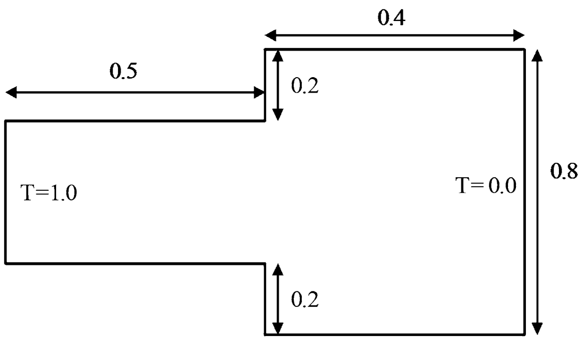


Fig. 3.13 Sketch of the problem 4

$T = 1.0$. Compare results obtained by LBM and FDM at time $= 50$ and 100. Hint: the relaxation time or relaxation frequency become function of temperature.

5. Anisotropic heat diffusion: heat or mass diffusion in wood, crystal, sedimentary rocks, and fibers have preferable direction. In other words, heat or mass diffusion coefficient is directional dependant. Heat diffusion equation in two dimensional domain can be expressed as,

$$\rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k_x \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k_y \frac{\partial T}{\partial y}\right) \tag{3.91}$$

In LBM modelling the relaxation time (or relaxation frequency) is directional dependent. The diffusion coefficients are related to relaxation frequency as,

$$\alpha_x = \frac{k_x}{\rho c} = \frac{\Delta x^2 c_s^2}{\Delta t}\left(\frac{1}{w_x} - 0.5\right) \tag{3.92}$$

and

$$\alpha_y = \frac{k_y}{\rho c} = \frac{\Delta y^2 c_s^2}{\Delta t}\left(\frac{1}{w_y} - 0.5\right) \tag{3.93}$$

c_s equal to $1/\sqrt{2}$ and $1/\sqrt{3}$ for D2Q4 and D2Q5, respectively. In calculating the distribution function that stream in x-direction w_x should be used and w_y should be used in calculating the distribution functions that stream in y-direction. Formulate the problem discussed in Sect. 3.10 for $\alpha_x = 0.5$ and $\alpha_y = 0.1$. Compare LBM and FDM predictions.